



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
PRE-SCORE
14-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S(TEAS)]

PAPER-1

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

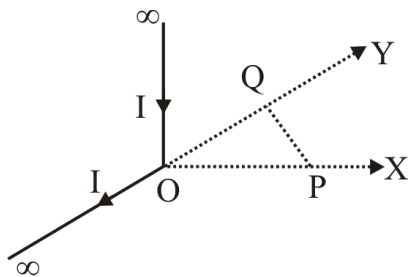
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Figure shows combination of two semi infinite long wires kept on Z and Y-axis. It carries constant current I as shown in figure. Calculate $\int_Q^P \vec{B} \cdot d\vec{\ell}$ on line QP. Co-ordinate of point P and Q are (a, 0, 0) and (0, a, 0) respectively.



(A) $\mu_0 I$

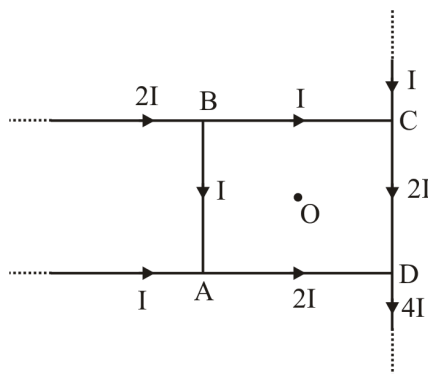
(B) $\frac{\mu_0 I}{2}$

(C) $\frac{\mu_0 I}{4}$

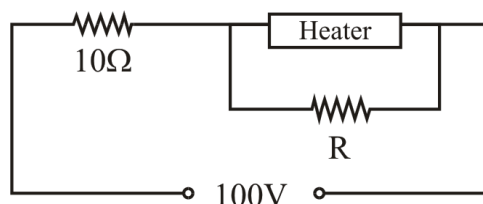
(D) $\frac{\mu_0 I}{8}$

Space for Rough Work

2. In the given circuit ABCD is a square of side 'a'. If magnetic field at point O is $\frac{N\mu_0 I(\sqrt{2} - 1)}{\sqrt{2}\pi a}$, find the value of N. Dotted line represents that wire goes upto infinity.



- (A) 2 (B) 4 (C) 6 (D) 3
3. A heater is designed to operate with a power of 1000 W in a 100 V line. It is connected in combination with a resistance of 10Ω and a resistance R to 100V mains as shown in figure. What should be the value of R so that the heater operates with a power of 62.5W?



- (A) 10Ω (B) 5Ω (C) 25Ω (D) 15Ω
4. A sphere of radius R carries charge such that its volume charge density is directly proportional to the square of the distance from the centre. What is the ratio of the magnitude of the electric field at a distance 2R from the centre to the magnitude of the electric field at a distance of R/2 from the centre (i.e. $\frac{E_{r=2R}}{E_{r=R/2}}$) ?
- (A) 4 (B) 2 (C) 3 (D) $2\sqrt{2}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

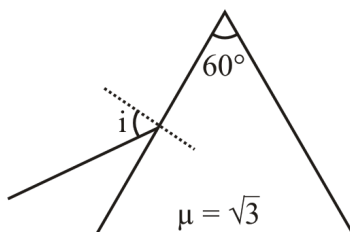
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

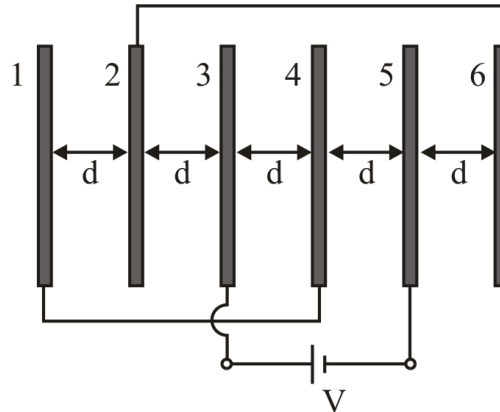
5. A plane polarized light strikes an equilateral prism ($\mu = \sqrt{3}$) kept in air. It is seen that there is no partial reflection from the first face. Choose the correct statement(s).



- (A) $i = 60^\circ$
- (B) The plane of electric field is the same as plane of incidence.
- (C) There is no reflected light from second face also.
- (D) The light ray undergoes minimum deviation.

Space for Rough Work

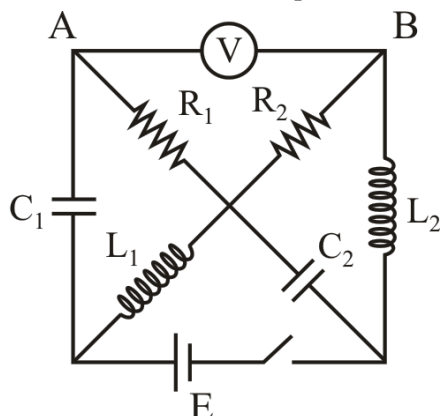
6. Consider an arrangement of six large parallel plates as shown in the figure, each of plate area A . Initially when battery was not connected all plates were uncharged. The emf of the battery is V . Choose the correct option(s) after connecting the battery (in steady state):



- (A) Total work done by battery is $\frac{\epsilon_0 A}{d} V^2$.
- (B) Electric field between plates 3 and 4 is $\frac{V}{2d}$.
- (C) Potential difference between plate 2 and 5 is $\frac{V}{2}$.
- (D) Potential difference between plate 1 and 5 is $\frac{V}{2}$.

Space for Rough Work

7. For the circuit shown in figure, $R_1 = 3R$, $R_2 = R$, $C_1 = C_2 = C$, and $L_1 = L_2 = L$. The electromotive force of the battery is E . Initially the switch is closed and the system is operating in a stationary regime. The voltmeter is ideal and it measures $V_A - V_B$. The switch is opened at $t = 0$. The reading of voltmeter :



- (A) Just before the opening switch is E .
- (B) Just after the opening of switch is $-2E$
- (C) Long time after opening the switch is E .
- (D) Is zero at any moment.
8. A positively charged particle with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The magnitude of magnetic force on particle is F . Which of the following statements are correct?
- (A) No force acts on particle if $x = y$.
- (B) $F \propto (x^2 - y^2)$ if $x > y$.
- (C) Force acts along z -axis if $x > y$.
- (D) Force acts along y -axis if $y > x$.

Space for Rough Work

9. Assume that we have a horizontal floor with coefficient of friction 0.2. There is no gravity, but an infinite vertical wire passes through the center of the floor. It carries a current of 80 A in the vertically upwards direction. A particle carrying charge of +5 mC and mass of 8 μ g is projected on the floor towards the wire from a distance of 4 m with a speed of 0.5 m/s. Choose the correct statement(s)
- (A) The charge particle will be lifted from the floor due to the force of the magnetic field.
- (B) The charge particle will be moving on the floor with decreasing speed.
- (C) The initial acceleration of the charge is 8 m/s².
- (D) The particle will stop after travelling $4 \left(1 - \frac{1}{\sqrt{e}} \right)$ m.
10. A particle having mass m and positive charge q whirls along vertical circle of radius R. At the centre of circle, another positive charge 2q is fixed.
- (A) The minimum velocity given at the lowest point to complete vertical circular motion is less than that for motion in gravity.
- (B) For any value of velocity given in horizontal direction at the lowest point in gravity free space it will perform vertical circular motion.
($\sqrt{5gR}$ is minimum velocity in case of gravity)
- (C) Tension in gravity free space cannot become zero at any point during motion.
- (D) If $mg < \frac{2kq^2}{R^2}$, the minimum speed at the lowest point so that it can complete vertical circle is $\sqrt{4gR}$.

Space for Rough Work

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

A fan operates at 200 volt (DC) consuming 1000 W when running at full speed. Its internal wiring has resistance 1Ω . When the fan runs at full speed, its speed becomes constant. This is because the torque due to magnetic field inside the fan is balanced by the torque due to air resistance on the blades of the fan and torque due to friction between the fixed part and the shaft of the fan. The electrical power going into the fan is spent (i) in the internal resistance as heat, call it P_1 (ii) in doing work against internal friction and air resistance producing heat, sound etc., call it P_2 . When the coil of fan rotates, an emf is also induced in the coil. This opposes the external emf applied to send the current into the fan. This emf is called back-emf. call it 'e'.

Answer the following questions when the fan is running at full speed.

1. The value of power P_1 (in W) is:
2. The value of power P_2 (in W) is:

Space for Rough Work

Paragraph for Questions 3 and 4

A transformer provides the last voltage stepdown in a power distribution system. A common sight on utility poles in residential areas, it is single-phase transformer that typically has a 12-kV rms line to neutral on its primary coil, and a secondary coil provides 240 V rms to service several residences. There are 10 houses connected to this transformer, each of which draw a current of 100 A rms. Assume a power factor of unity. Consider the situation illustrated in figure, showing two coils of wire wound around a single closed magnetic core. Assume a core flux ϕ , which links all the turns of both coils. In the ideal case, we also neglect wire resistance. Let us now examine the coupling equations under the condition that the same flux goes through each winding, and so

$$v_1(t) = N_1 \frac{d\phi}{dt} \text{ and } v_2(t) = N_2 \frac{d\phi}{dt}$$

and therefore, $\frac{v_1}{v_2} = \frac{N_1}{N_2} \frac{\frac{d\phi}{dt}}{\frac{d\phi}{dt}} = \frac{N_1}{N_2} \dots(i)$

Ampere's law requires that

$$\oint \mathbf{H} \cdot d\mathbf{l} = i_{\text{enclosed}} = N_1 i_1 + N_2 i_2 \dots(ii)$$

where H is the magnetic field intensity and the integral is over the closed path travelled by the flux around the transformer core. If $H = 0$, which is the case for an ideal for an ideal magnetic core with infinite permeability, then

$$N_1 i_1 + N_2 i_2 = 0 \dots(iii)$$

$$\text{or } \frac{i_1}{i_2} = -\frac{N_2}{N_1} \dots(iv)$$

Note that if we divide Eq. (iii) by N_1 and multiply it by v_1 , we obtain

$$v_1 i_1 + \frac{N_2}{N_1} v_1 i_2 = 0$$

$$\text{However, since } v_1 = \left(\frac{N_1}{N_2} \right) v_2.$$

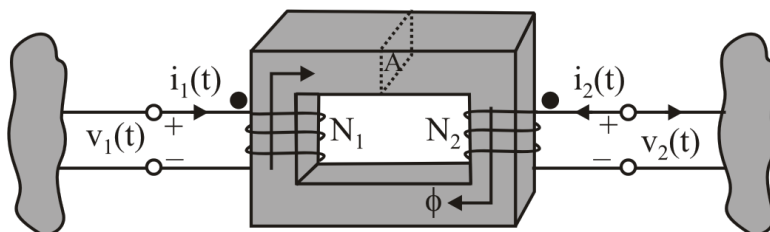
$$v_1 i_1 + v_2 i_2 = 0$$

and hence the total power into the device is zero, which means that an ideal transformer is lossless.

The symbol we employ for the ideal transformer is shown in figure and the corresponding equations are

$$\frac{v_1}{v_2} = \frac{N_1}{N_2}$$

$$N_1 i_1 + N_2 i_2 = 0 \dots(v)$$

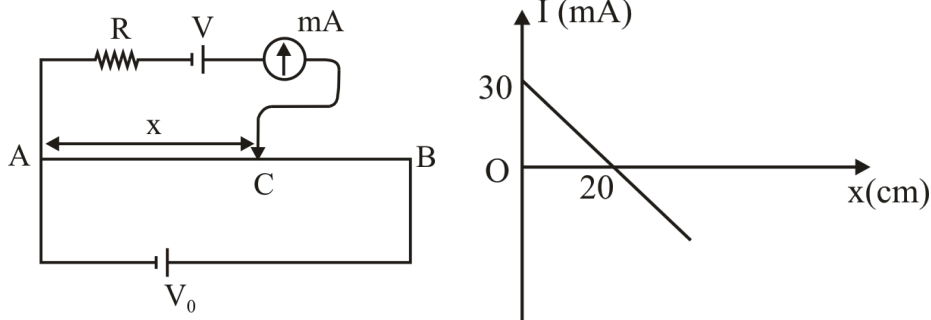


Transformer employing a magnetic core

3. What is the ratio of the number of turns in primary and secondary indicated in the passage?
4. What is the ratio of self-inductance of primary and mutual inductance of primary and secondary?

Paragraph for Questions 5 and 6

AB is a uniform wire of length $L = 100$ cm. A cell of emf $V_0 = 12$ volt is connected across AB. A resistance R , cell of emf V and a milliammeter (which can show deflection in both directions) is connected to the circuit as shown. Contact C can be slid on the wire AB. Distance $AC = x$. The current (I) through the milliammeter is taken positive when the cell of emf V is discharging. A graph of I Vs x has been shown. Neglect internal resistance of the cells.



5. Find V (in volts)
6. Find resistance R in ohms.

Space for Rough Work

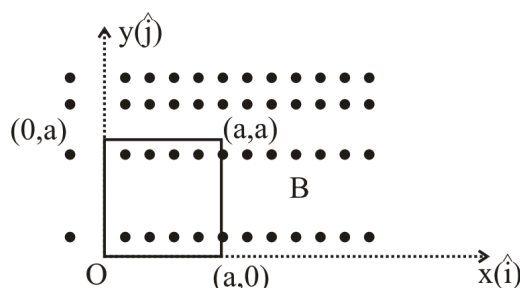
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

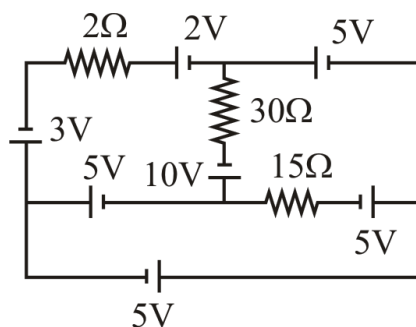
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. A single degree of freedom viscous damped system of mass 10 kg has a spring stiffness of 6000 N/m, damping constant 0.3 Ns/mm. If the system is given an initial velocity of $\sqrt{15}$ m/sec, determine the initial amplitude (in cm) of the system.
8. A square loop of side 'a' is placed in x - y plane as shown in figure. In this region there is non-uniform time dependent magnetic field $\vec{B} = (cy^3t^2)\hat{k}$. [where t is time and c is constant]. The magnitude of emf induced in loop is $\frac{Ca^5t}{P}$. Find the value of P.



9. In the circuit shown, current through the resistance 2Ω is i_1 and current through the resistance 30Ω is i_2 . Find the ratio $\frac{i_1}{i_2}$.



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

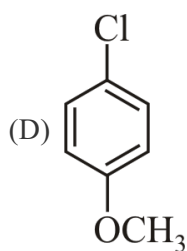
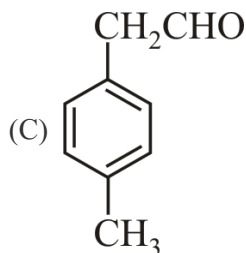
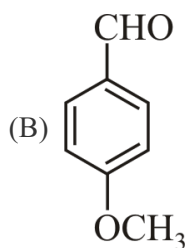
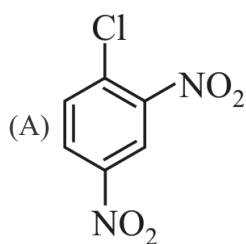
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

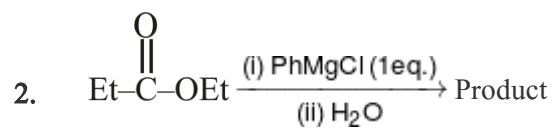
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. The compound(s) that will NOT react(s) with hot concentrated aqueous alkali at atmospheric pressure is / are :

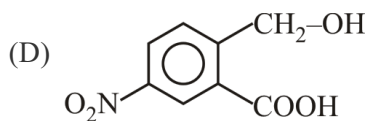
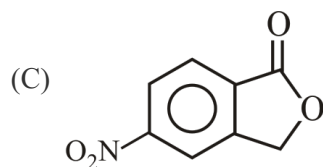
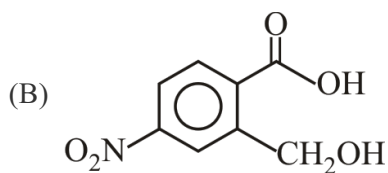
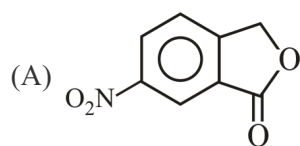
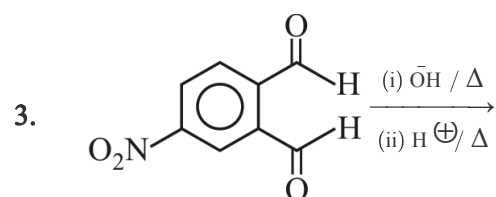


Space for Rough Work



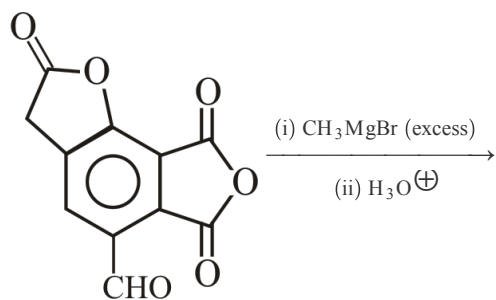
True about product of above reaction :

- (A) One of the product will give positive iodoform test
- (B) Product will give CO_2 on heating
- (C) Product will give only one oxime with NH_2OH
- (D) Product will give chiral alcohol with EtMgBr followed by hydrolysis.

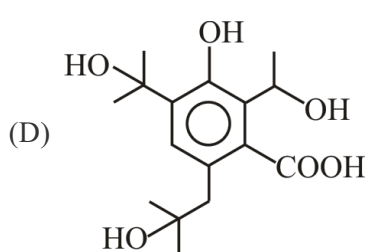
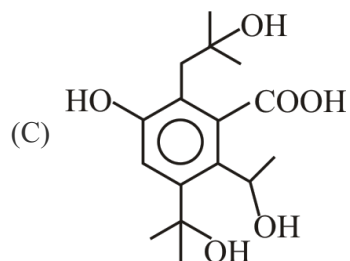
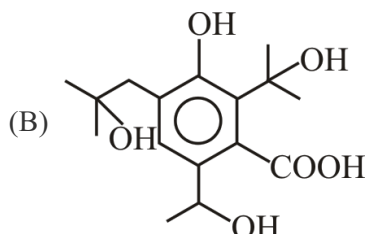
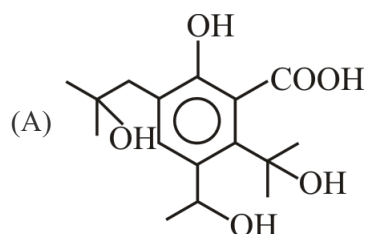


Space for Rough Work

4.



Major product is



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

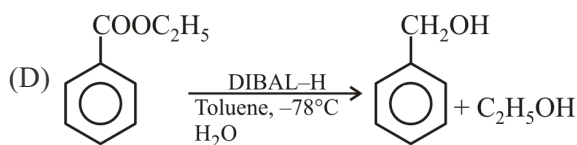
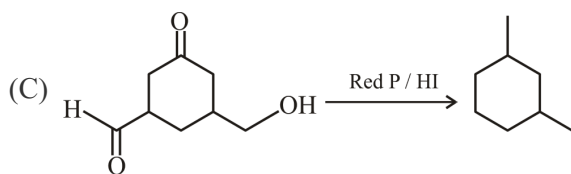
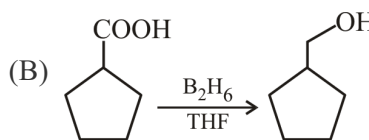
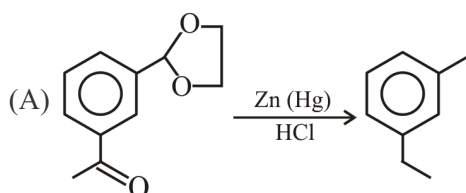
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Which of the following represents correct product formation?



6. Which of the following will give CO_2 gas with aq. NaHCO_3 and contains $-\text{COOH}$ group?

(A) Tartaric acid

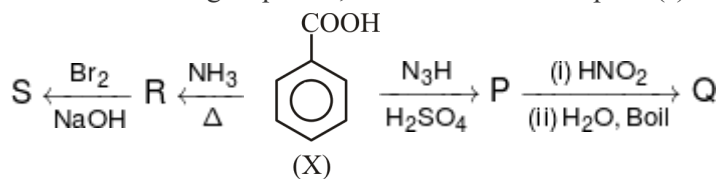
(B) Picric acid

(C) Aspirin

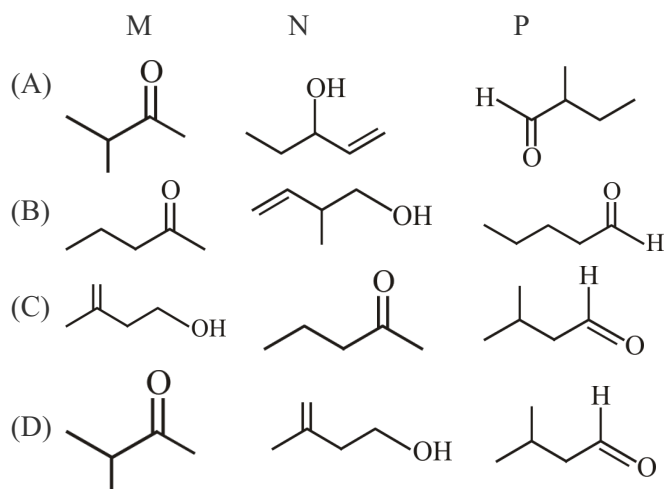
(D) Aspartic acid

Space for Rough Work

7. For the following sequence, choose the correct option(s)

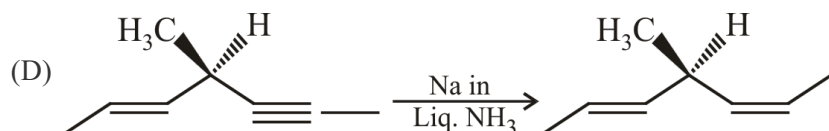
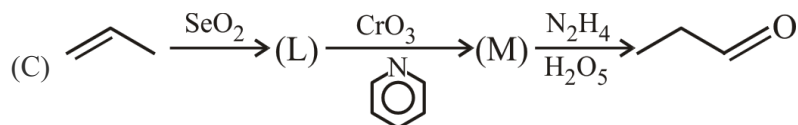
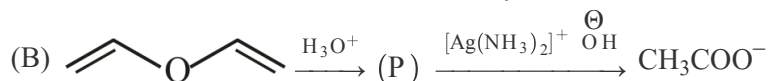
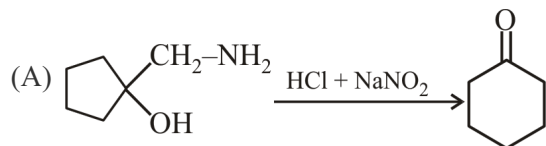


- (A) (Q) will give positive Br_2 , H_2O test
- (B) (R) on reaction with NaNO_2 , HCl produces (X)
- (C) Major product (P) and (S) are identical
- (D) Reaction sequence involves schmidt and Hoffmann bromamide degradation reaction
8. Three isomeric compound M, N and P ($\text{C}_5\text{H}_{10}\text{O}$) give the following tests :
- (i) M and P react with sodium bisulfite to form an adduct
- (ii) N consumes 1 mole of bromine and also gives turbidity with conc. HCl / anhydrous ZnCl_2 after prolong heating
- (iii) M react with excess of iodine in alkaline solution to give yellow crystalline compound with a characteristic smell.
- (iv) p-Rosaniline treated with sulphur dioxide develops pink colour on shaking with P.
- The structures of M, N and P, respectively can be :

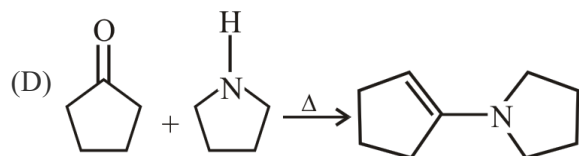
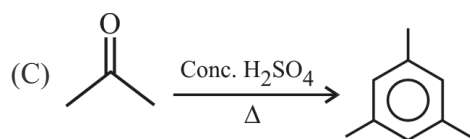
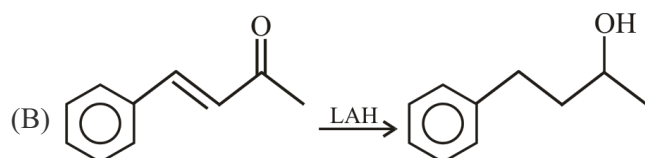


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9. Which of the given sequence are correctly matched with their final major product :



10. Select reaction(s) correctly matched with major product :



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SECTION-II (i) : (Maximum Marks: 12)

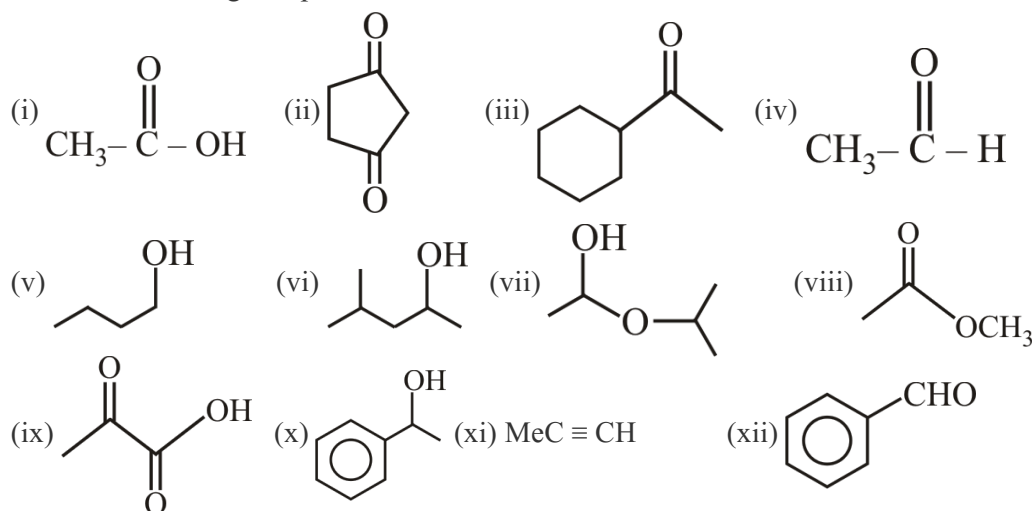
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider following compounds



- How many of the given show positive Iodoform test?
- How many of the given compounds give positive silver mirror test with Tollen's reagent

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Paragraph for Questions 3 and 4

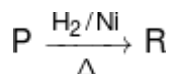
Consider all possible compounds of molecular mass 100 which gives yellow colour with 2,4-DNP. All these isomers are independently reacted with LiAlH_4 . Answer following question based on the observation.

3. Product of how many can be used to resolve d and l 2-methylbutanoic acid?
4. How many of the will give optically inactive product?

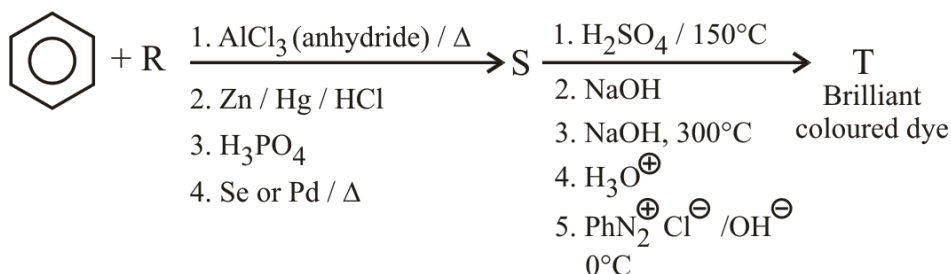
Paragraph for Questions 5 and 6

P and Q are isomer or dicarboxylic acid $\text{C}_4\text{H}_4\text{O}_4$ both decolorized $\text{Br}_2 / \text{H}_2\text{O}$. On heating, P forms the cyclic anhydride. Upon treatment with dilute alkaline KMnO_4 , P as well as Q could produce superimposable on its mirror image in one isomer and non-superimposable on its mirror- image in other isomers.

5. In the following reaction sequence (R) has molecular mass



6. Degree unsaturation of product T is :



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SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

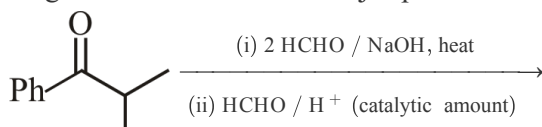
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

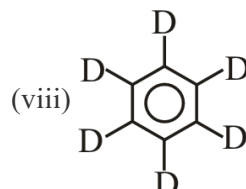
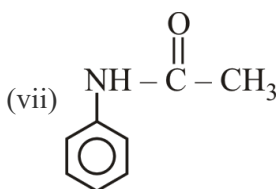
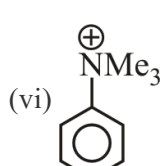
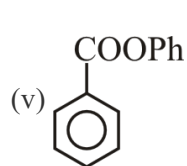
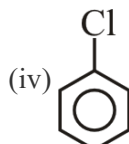
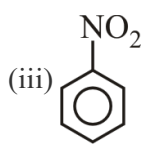
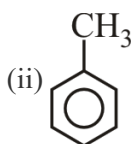
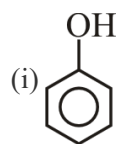
7. Total number of stereoisomers of following compound which contains plane of symmetry is :



8. Degree of unsaturation of major product obtained in following reaction is :



9. How many of following compounds are less reactive than benzene for sulphonation by conc. H_2SO_4



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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. PQ is a chord of parabola $x^2 = 4y$ which subtends right angle at vertex. Then locus of centroid of triangle PSQ, where S is the focus of given parabola, is a conic having focus at

(A) $\left(0, \frac{1}{3}\right)$	(B) $\left(0, \frac{10}{3}\right)$
(C) $\left(0, \frac{8}{3}\right)$	(D) none of these

2. For an ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ with vertices A and A', tangent drawn at the point P in the first quadrant meets the y-axis in Q and the chord A'P meets the y-axis in M. If 'O' is the origin then $OQ^2 - MQ^2$ equals to :

(A) 2	(B) 6	(C) 3	(D) 4
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Space for Rough Work

3. L_1 is a tangent drawn to the curve $x^2 - 4y^2 = 16$ at $A \left(5, \frac{3}{2} \right)$. L_2 is another tangent parallel to L_1 which meets the curve at B. L_3 and L_4 are normals to the curve at A and B. Lines L_1, L_2, L_3, L_4 forms a rectangle, then which of the following is correct
- (A) Equation of tangent at B is $6y = 5x - 16$.
- (B) Equation of normal at B is $12x + 10y + 74 = 0$
- (C) Radius of largest circle inscribed in the rectangle is $\frac{32}{\sqrt{61}}$.
- (D) Radius of the circle circumscribing the rectangle is $\frac{\sqrt{109}}{2}$
4. If λ_1 is the number of integers in the solution set of $|\cos 3x| + |\cos x| = |\cos 3x + \cos x|$ that belongs to $[0, \pi]$ and λ_2 is the tangent of angle between the tangents drawn from origin to the parabola $y^2 = 4(x - 4)$, then $(\lambda_1 + \lambda_2)$ equals
- (A) $\frac{10}{3}$ (B) $\frac{11}{3}$
- (C) $\frac{13}{3}$ (D) 4

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
5. A variable chord of $\frac{x^2}{9} - \frac{y^2}{b^2} = 1$ subtends 90° at the centre of the hyperbola. If this chord touches a fixed circle concentric with the hyperbola then

(A) radius of circle is an irrational number for $b = 2$

(B) radius of circle is a rational number for $b = 5$

(C) centre of circle is at (0,0)

(D) the circle passes through $\left(\sqrt{3}, \sqrt{\frac{123}{7}}\right)$ for $b = 4$

Space for Rough Work

6. The sides of ΔABC are tangents to parabola $y^2 = 8x$. Sides BC, CA and AB touch the parabola at D, E and F respectively. If lines AD, BE and CF are concurrent at focus (S) of parabola, such that $BD = 8\sqrt{3}$, $D(t_1)$, $E(t_2)$, $F(t_3)$, $t_1 > t_2 > t_3$, then which of the following option is/are correct ?
- (A) $t_1^2 + t_2^2 + t_3^2 = 6$ (B) $CF + AF = 6\sqrt{3}$
 (C) $AE \cdot CD = 12$ (D) $t_1 t_2 + t_2 t_3 + t_3 t_1 = 3$
7. A parabola passes through A (3,7) and B (5,11). If line AB intersects the directrix of parabola at T(1,3), and $ax^2 + 2hxy + by^2 + 2gx + 2fy + 526 = 0$ represents the locus of focus of parabola, then which of the following options is/are correct?
- (A) $20a + g = 43$ (B) $20a + g = 53$
 (C) $30b + 45h + f = 73$ (D) $30b + 45h + f = 63$
8. Consider an ellipse E : $\frac{x^2}{16} + \frac{y^2}{4} = 1$
- (A) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $4x^2 + y^2 = 9$ (where O is the origin)
 (B) Normal is drawn at a point P on E to intersect x and y axis at A and B respectively then locus of circumcentre of ΔOAB is $x^2 + 9y^2 = 4$ (where O is the origin)
 (C) Area of quadrilateral formed by foci and end points of minor axes of the given ellipse is $8\sqrt{3}$ sq. units
 (D) If a tangent is drawn at P on E to intersect the x-axis and y-axis at L and M respectively then minimum value of LM is 6 units

Space for Rough Work

9. Consider an ellipse with major axis and minor axis having length 4 and $2\sqrt{3}$ respectively. Let P be variable point on it and S and S' be the foci and C be the centre of ellipse then
- (A) maximum value of $PS \cdot PS'$ is 4
- (B) minimum value of $PS^2 + PS'^2$ is 8
- (C) minimum length of portion of tangent at P intercepted between the principal axes is $\sqrt{7 - 4\sqrt{3}}$
- (D) If the foot of perpendicular from centre upon the normal at point P is the point Q then the maximum value of PQ is $2 - \sqrt{3}$
10. If point (4, 4) lies on parabola whose focus lies on x-axis and directrix is $\ell x + my = 1$ such that (ℓ, m) lies on curve $x^2 + y^2 - 32xy + 8x + 8y - 1 = 0$ then :
- (A) co-ordinates of focus may be (5, 0)
- (B) co-ordinates of focus may be (3, 0)
- (C) If tangent at (4, 4) passes through origin, value of m may be $\frac{1}{3}$
- (D) If tangent at (4, 4) passes through origin, equation of axis may be $x - 4y - 3 = 0$

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SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Let a chord of the parabola $y^2 = 4x$ subtend a right angle at the centre of the circle $x^2 + y^2 = 16$ and the chord passes through a fixed point P and from point P three normal are drawn to intersect the parabola at A, B, C.

1. If area of $\triangle ABC$ is k, then value k^2 is
2. Out of the 3 normal chord, the minimum length of normal chord is L, then L^2 is

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Paragraph for Questions 3 and 4

The circle $C_1 : x^2 + y^2 = r^2$ and ellipse $C_2 : \frac{x^2}{16} + \frac{y^2}{9} = 1$ intersect at 4 distinct points A,B,C,D. Their common tangents form a parallelogram A'B'C'D'.

3. If A'B'C'D' is a square then r is equal to $\frac{a}{\sqrt{b}}$ where a and b are coprime then value of (a + 2b) is -
4. If A'B'C'D' is a square, then the ratio of area of the circle C_1 to the area of the circumcircle of $\Delta A'B'C'$ is equal to

Paragraph for Questions 5 and 6

Consider the hyperbola $y = x - \frac{1}{x}$.

5. If eccentricity of hyperbola is $\sqrt{2a + a\sqrt{a}}$, then value of 100a is
6. The length of its semi-transverse axis is $\sqrt{2\sqrt{b} - 2}$, then value of 50b is

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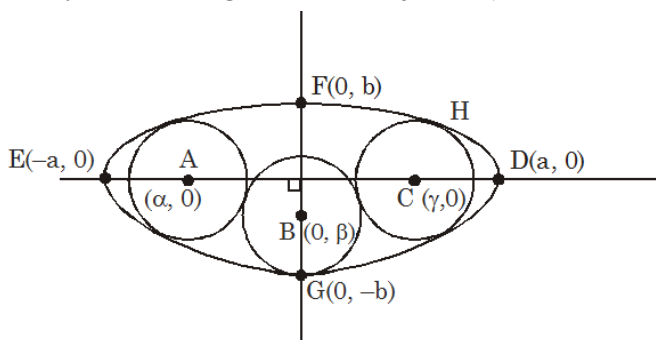
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Tangents are drawn from any point on the hyperbola $4x^2 - 9y^2 = 36$ to the circle $x^2 + y^2 - 9 = 0$. The locus of mid-point of chord of contact is $\left(\frac{x^2}{9} - \frac{y^2}{4}\right) = \left(\frac{x^2 + y^2}{k}\right)^2$, where $k \in \mathbb{N}$. Find k .
8. Three identical unit circles are packed in an ellipse, as shown in the figure. Let A, B, C be the centre of these circles and E, D are the vertices of major axis and F, G are vertices of minor axis of ellipse. If the angle $\angle ABC$ such that the area of the ellipse is minimized is $2\tan^{-1}\sqrt{k}$, then the value of $\frac{63e^2a^2}{2k}$ is equal to (where e is eccentricity and a is length of semi-major axis)



9. Consider the equation of an ellipse E and lines L, M, N
- $$E \equiv \frac{(x-4)^2}{16} + \frac{(y-3)^2}{9} = 1, \begin{pmatrix} L \equiv 3x + 3y = 4 \\ M \equiv 3y = 3x - 2 \\ N \equiv 4x = 4y + 3 \end{pmatrix}.$$

The reflection of ellipse E, in line L is ellipse F. The reflection of ellipse F in line M is another ellipse G. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents the equation of ellipse, who is a reflection of ellipse G in line N, then $\frac{9ag + 152bf}{bg}$ is equal to k then value of $2k$ is

Space for Rough Work

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

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